CHA-3.D.1 The chain rule is the basis for differentiating variables in a related rates problem with respect to the same independent variable.

CHA-3.D.2 Other differentiation rules, such as the product rule and the quotient rule, may also be necessary to differentiate all variables with respect to the same independent variable.

CHA-3.E.1 The derivative can be used to solve related rates problems; that is, finding a rate at which one quantity is changing by relating it to other quantities whose rates of change are known.

4-4 & 4-5 Related Rates

Get Ready...

Consider the curve defined by $x^2 - 2y + 5x = y^3 - 6$. Find $\frac{dy}{dx}$.

$$2x - 2\frac{dy}{dx} + 5 = 3y^2\frac{dy}{dx}$$

$$2x + 5 = 3y^2\frac{dy}{dx} + 2\frac{dy}{dx}$$

$$2x + 5 = \frac{dy}{dx}\left(3y^2 + 2\right)$$

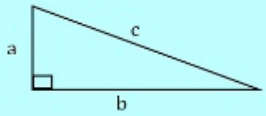
$$\frac{2x+5}{3y^2+2} = \frac{dy}{dx}$$

In the example above, we found $\frac{dy}{dx}$, which means we differentiated **with respect to x**. However, we can differentiate an equation with respect to **ANY** variable, even if that variable is **NOT** in the equation. We do this in the same way we do implicit differentiation with the chain rule.

In math, two variables are related if one variable affects the other. We see this with various formulas.

<u>Triangle</u>

Let's say a triangle's dimensions are changing. This means a dimension (a, b, or c) has a rate of change, and this rate is "related" to another dimension's rate of change. What equation relates the variables a, b, and c ?



$$a^2+b^2=c^2$$

Just as values can be related through equations, the derivatives of those values are also related. Finding how the rates are related requires differentiation with respect to a given variable, usually with respect to time. We will use implicit differentiation to find the derivative. What is the derivative of our triangle equation above, with respect to t, time?

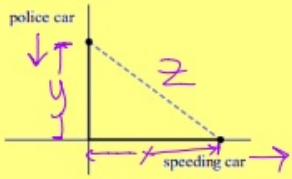
$$a^2+b^2=c^2$$

$$2a\cdot\frac{da}{dt}+2b\frac{db}{dt}=2c\cdot\frac{dc}{dt}$$

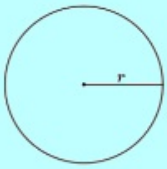
Example 1: A police car, approaching a right-angled intersection from the north, is chasing a speeding car that has turned the corner and is now moving straight east. Set up a relationship, then find an equation that shows the related rates of the vehicles with respect to time.

$$x^2 + y^2 = z^2$$

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 2z \cdot \frac{dz}{dt}$$



Circle



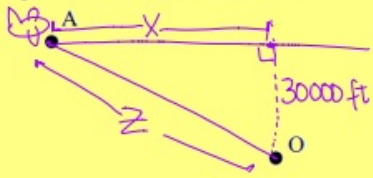
Consider the circle at the right with radius r. We know that the radius of the circle is related to the area of the circle. When r changes, we know the area of the circle also changes. The area and radius of the circle are related by the equation $A = \pi r^2$. Find the derivative of the area equation, with respect to time.

$$A = \pi r^2$$

$$\frac{dA}{dt} = \pi \cdot 2r \cdot \frac{dr}{dt}$$

$$\frac{dA}{dt} = 2\pi r \cdot \frac{dr}{dt}$$

Example 2: An airplane (point A) is flying on a horizontal path that will take it directly over an observer (point O). The airplane maintains a constant altitude of 30,000 feet. Relate the rates of change of the distance between the observer and the airplane and the horizontal distance between the two.



$$x^2 + (30000)^2 = z^2$$

$$2x\frac{dx}{dt} = 2z\frac{dz}{dt}$$

YOU TRY A: A particle is moving in circular orbit $x^2 + y^2 = 25$. As it passes through the point $(3, 4)$ its y-coordinate is decreasing at the rate of 2 m/s. How is the x-coordinate changing?

$x^2 + y^2 = 25$

$(3,4) \rightarrow (x,y)$

$x = 3 \quad y = 4$

$\frac{dy}{dt} = -2 \text{ m/s} \quad \frac{dx}{dt} = ?$

$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$

$x\frac{dx}{dt} + y\frac{dy}{dt} = 0$

$3\frac{dx}{dt} + 4(-2) = 0$

$3\frac{dx}{dt} - 8 = 0$

$3\frac{dx}{dt} = 8$

$\frac{dx}{dt} = \frac{8}{3} \text{ m/s}$

the x-coordinate is increasing at a rate of $\frac{8}{3}$ m/s

To solve related rates problems, you need a strategy that always works. Related rates problems always can be recognized by the words "**increasing, decreasing, growing, shrinking, changing**." Follow these guidelines in solving a related rates problem.

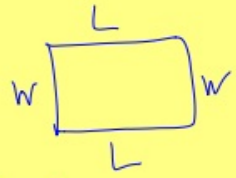
Guidelines for Solving Related Rate Problems
Step 1: Make a sketch. Label all sides in terms of variables even if you are given the actual values of the sides.
Step 2: Write down the sides and the rates that are given **with the correct sign (positive or negative).** **If a quantity is increasing, its derivative is positive. If a quantity is decreasing, its derivative is negative.** Also, write down the rate you are trying to find.
Step 3: Find an equation that ties your variables together. • If the problem mentions **area**, you will most likely write an **area formula**. • If the problem mentions **volume**, you will most likely write a **volume formula**. • If the problem mentions lengths of sides and **the figure is a right triangle**, you will most likely use the **Pythagorean Theorem** as your equation. • If the problem mentions angles measures and **the figure is a right triangle**, you will most likely use a **trigonometric ratio** as your equation.
Step 4: You may now plug in any constant into your equation. **Never** plug in any variable into your equation until **after** the derivative is taken.
Step 5: Differentiate your equation with respect to time. Remember, you are implicitly differentiating with respect to t.
Step 6: Plug in all variable quantities mentioned in the "given" and "when" parts of the problem. Hopefully you will know all the variables except for one. If not, you will need an equation which will solve for unknown variables. Sometimes this will require using the same equation as the one you used above. Do this work off to the side.
Step 7: Label your answers in terms of the correct units (very important) and be sure you answered the question asked.

Example 3: The width of a rectangle is increasing at a rate of 2 cm/sec and its length is increasing at a rate of 3 cm/sec. At what rate is the area of the rectangle increasing when its width is 4 cm and its length is 5 cm?

Given: $\frac{dw}{dt} = 2\text{cm/sec}$

$\frac{dL}{dt} = 3\text{cm/sec}$

Picture:



Find: $\frac{dA}{dt} = ?$ when $W = 4\text{cm}$ and $L = 5\text{cm}$

Equation: $A = L \cdot W$

$$\frac{dA}{dt} = \frac{dL}{dt} \cdot W + L \cdot \frac{dW}{dt}$$

$$\frac{dA}{dt} = 3(4) + 5(2)$$

$$\frac{dA}{dt} = 22 \text{ cm}^2/\text{s}$$

the area of the rectangle is increasing at a rate of $22\text{cm}^2/\text{s}$

Example 4: Air is being pumped into a spherical balloon so that its volume increases at a rate of 100 cm³/s. How fast is the radius of the balloon increasing when the radius is 25 cm?

Given: $\frac{dV}{dt} = 100 \text{cm}^3/\text{s}$

Picture:

Find: $\frac{dr}{dt} = ?$ When $r = 25\text{cm}$

Equation: $V = \frac{4}{3}\pi r^3$

$$\frac{dV}{dt} = \frac{4}{3}\pi \cdot 3r^2 \cdot \frac{dr}{dt}$$

$$\frac{dV}{dt} = 4\pi r^2 \cdot \frac{dr}{dt}$$

$$100 = 4\pi (25)^2 \cdot \frac{dr}{dt}$$

$$100 = 2500\pi \cdot \frac{dr}{dt}$$

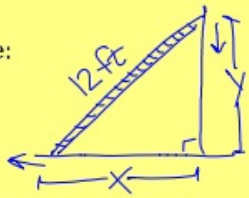
$$\frac{dr}{dt} = \frac{100}{2500\pi} = \frac{1}{25\pi} \text{ cm/s}$$

The radius of the balloon is increasing at a rate of $\frac{1}{25\pi}$ cm/s

Example 5: A 12-foot ladder is leaning up against a wall when it begins to slide down the wall at a rate of 2 ft/sec. At what rate is the base of the ladder moving away from the wall when the ~~it~~ ladder is 5 feet from the wall?

Given: $\frac{dy}{dt} = -2 \text{ ft/s}$

Picture:



Find: $\frac{dx}{dt} = ?$ when $x = 5 \text{ ft}$

Equation: $x^2 + y^2 = 12^2$

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

$$x\frac{dx}{dt} + y\frac{dy}{dt} = 0$$

$$5\frac{dx}{dt} + \sqrt{119}(-2) = 0$$

$$5\frac{dx}{dt} = 2\sqrt{119}$$

$$\frac{dx}{dt} = \frac{2\sqrt{119}}{5} \text{ ft/s}$$

$$x^2 + y^2 = 12^2$$

$$5^2 + y^2 = 12^2$$

$$y^2 = 119$$

$$y = \sqrt{119}$$

the base of the ladder is moving away from the wall at a rate of $\frac{2\sqrt{119}}{5}$ ft/s

(4.363 ft/s)

Example 6: Water is being poured into a cylindrical container at a rate of 4 cubic meters per second. The radius of the jar is 3 meters. At what rate is the height of the water changing when the water is at a height of 10 meters?

Given: $\frac{dV}{dt} = 4\ m^3/s$

Picture:

Find: $\frac{dh}{dt} = ?$ when $h = 10m$

Equation: $V = \pi(3)^2 h$

$V_{cyl} = \pi r^2 h$

$V = 9\pi h$

$\frac{dV}{dt} = 9\pi \cdot \frac{dh}{dt}$

$4 = 9\pi \cdot \frac{dh}{dt}$

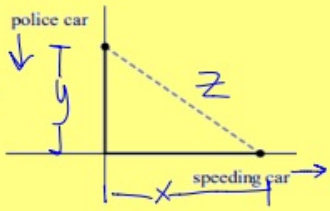
$\frac{dh}{dt} = \frac{4}{9\pi}\ m/s$

The height of the water is increasing at a rate of $\frac{4}{9\pi}$ m/s

Example 7: A police car, approaching a right-angled intersection from the north is chasing a speeding car that has turned the corner and is now moving straight east. When the police car is 0.6 miles north of the intersection and the car is 0.8 miles to the east, the police determine with a radar gun that the distance between them and the car is increasing at 20 mph. If the police car is moving at 60 mph at the instant of measurement, what is the speed of the car?

When $y = 0.6$ mi, $x = 0.8$ mi, $\frac{dz}{dt} = 20 \frac{mi}{hr}$

$\frac{dy}{dt} = -60$ mi/hr, $\frac{dx}{dt} = ?$



$$x^2 + y^2 = z^2$$

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 2z\frac{dz}{dt}$$

$$x\frac{dx}{dt} + y\frac{dy}{dt} = z\frac{dz}{dt}$$

$$0.8\frac{dx}{dt} + 0.6(-60) = 1(20)$$

$$0.8\frac{dx}{dt} = 56$$

$$\frac{dx}{dt} = 70 \text{ mi/h}$$

$$x^2 + y^2 = z^2$$

$$0.8^2 + 0.6^2 = z^2$$

$$1 = z^2$$

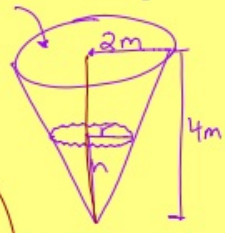
$$1 = z$$

the car is moving away from the intersection at a rate of 70mi/hr.

Cone Problem (Similar Triangles)

Example 8: A water tank has the shape of an inverted circular cone with a base radius 2 m and height 4 m. If water is being pumped into the tank at a rate of 2 m^3/min, find the rate at which the water level is rising when the water is 3 m deep. The volume of a cone is $V = \frac{1}{3}\pi r^2 h$.

$\frac{dV}{dt} = 2\ m^3/min$ $\quad \frac{dh}{dt} = ?$ when $h = 3m$



$$V = \frac{1}{3}\pi r^2 h$$

$$\frac{r}{h} = \frac{2}{4}$$

$$\frac{4r}{4} = \frac{2h}{4}$$

$$r = \frac{1}{2}h$$

$$V = \frac{1}{3}\pi\left(\frac{1}{2}h\right)^2 h$$

$$V = \frac{1}{12}\pi h^3$$

$$\frac{dV}{dt} = \frac{1}{12}\pi \cdot 3h^2 \cdot \frac{dh}{dt}$$

$$\frac{dV}{dt} = \frac{1}{4}\pi h^2 \frac{dh}{dt}$$

$$2 = \frac{1}{4}\pi(3)^2 \frac{dh}{dt}$$

$$\frac{4}{9\pi} \cdot 2 = \frac{9\pi}{4}\frac{dh}{dt} \cdot \frac{4}{9\pi}$$

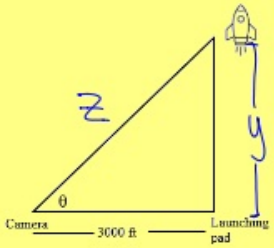
$$\frac{dh}{dt} = \frac{8}{9\pi}\ m/min$$

The water level is rising at a rate of $\frac{8}{9\pi}$ m/min

Example 9: A camera is mounted at a point 3000 ft from the base of a rocket launching pad. If the rocket is rising vertically at 880 ft/s, when it is 4000 ft above the launching pad, how fast must the camera elevation angle change at that instant to keep the camera aimed at the rocket?

$\frac{dy}{dt} = 880 \text{ ft/s}$

When $y = 4000$ ft $\quad \frac{d\theta}{dt} = ?$



$$\tan\theta = \frac{y}{3000}$$

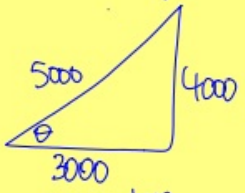
$$\sec^2\theta \cdot \frac{d\theta}{dt} = \frac{1}{3000} \cdot \frac{dy}{dt}$$

$$\left(\frac{5}{3}\right)^2 \cdot \frac{d\theta}{dt} = \frac{1}{3000}(880)$$

$$\frac{9}{25} \cdot \frac{25}{9} \frac{d\theta}{dt} = \frac{880}{3000} \cdot \frac{9}{25}$$

$$\frac{d\theta}{dt} = \frac{66}{625} \text{ radians/sec}$$

When $y = 4000$



$\sec\theta = \frac{\text{hyp}}{\text{adj}}$

$\sec\theta = \frac{5000}{3000} = \frac{5}{3}$

The camera elevation angle is increasing at a rate of $\frac{66}{625}$ rad/sec

Shadow problem:

Tips for shadow problems:

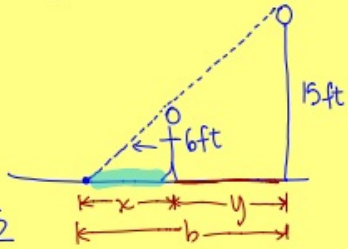
The rate at which the shadow changes is the rate of change of the base of the small triangle.

The rate at which the tip of the shadow changes is the rate of change of the base of the large triangle.

Example 10: A man 6 feet tall walks at a rate of 5 feet per second away from a light that is 15 feet above the ground. When he is 10 feet from the base of the light, at what rate is the tip of his shadow changing?

$\frac{dy}{dt} = 5$ ft/sec

When $y = 10$ ft $\quad \frac{db}{dt} = ?$



$$x + y = b$$

$$\frac{6}{x} = \frac{15}{b}$$

$$6b = 15x$$

$$\frac{6}{15}b = x$$

$$\frac{2}{5}b = x$$

$$\frac{2}{5}b + y = b$$

$$-\frac{2}{5}b \qquad -\frac{2}{5}b$$

$$y = \frac{3}{5}b$$

$$\frac{dy}{dt} = \frac{3}{5}\frac{db}{dt}$$

$$\frac{5}{3} \cdot 5 = \frac{3}{5}\frac{db}{dt} \cdot \frac{5}{3}$$

$$\frac{25}{3} \frac{ft}{sec} = \frac{db}{dt}$$

The tip of his shadow increasing at a rate of $\frac{25}{3}$ ft/sec